

CHEMISTRY & BIOCHEMISTRY

Courses

Below you will find a list of our current or recent offerings. See the [course catalog](#) for descriptions and updated information.

51-014 Chemistry Appreciation

A presentation of historic and modern theories and concepts of the nature of matter and bonding. Current problems dealing with synthetic and natural products and their pollutants will be discussed in light of their impact on society. Individual experiments and group demonstrations will be conducted in the laboratory. Contributes to Health Studies. (NS)

51-024 Chemistry of the Environment

This course provides an overview of basic chemical principles and their importance in understanding the complexities of our natural environment. In particular, the course will discuss fundamental chemical concepts such as equilibrium, solubility and acid-base chemistry and their application to environmental processes. Major topics that will be covered include atmospheric and aquatic chemistry, energy production and usage, and principles of toxicology. Contributes to Environmental Studies. (NS)

51-034 Chemistry of Food

Students in this course will develop a better understanding of food, cooking, and nutrition using basic chemical concepts. Topics that will be discussed include the impact our food choices make on the environment; modern agricultural practices; and the economic, political and social justice issues surrounding the use of food and its availability. Contributes to Environmental Studies. (NS)

51-080 Fundamentals of Chemistry Recitation

 Menu Search[Apply](#) [Visit](#) [Majors & Minors](#)

their understanding through collaborative problem solving and group work.

51-083 Fundamentals of Chemistry

This course is designed to strengthen the foundational knowledge and problem-solving skills essential for success in General Chemistry I and II. Emphasis is placed on building confidence with key concepts such as atomic structure, periodic trends, chemical bonding, stoichiometry, and thermochemistry. Students will develop strategies for approaching multi-step problems, interpreting symbolic and graphical representations, and understanding the language of chemistry. The course also fosters scientific reasoning and critical thinking through guided practice and application-based exercises. (NS)

51-100 General Chemistry Recitation

This course provides students with additional time to practice what they have learned in the General Chemistry I or II lecture. Students will discuss topics, apply concepts, and assess their understanding through collaborative problem solving and group work. May be repeated.

51-101 Chemical Methods & Techniques Lab I

This laboratory course will provide students an opportunity to learn and practice common laboratory techniques through self-directed laboratory experiments. Contributes to Health Studies. To be taken concurrently with Chemistry 51-103. (NS) (Fall)

51-103 General Chemistry I

This course will introduce students to fundamental chemical principles and concepts such as atomic structure, chemical bonding, stoichiometry, thermochemistry, periodicity, solution chemistry, properties of gases and selected topics in descriptive chemistry. Contributes to Health Studies. Prerequisite: Mastery of high school-level chemistry and math. Students will need to complete a Chemistry Readiness Assessment as a prerequisite for the course. To be taken concurrently with Chemistry 51-103. (NS) (Fall)

51-111 Problem Solving in Chemistry

Students in this course will learn and apply systematic approaches to solving problems

 Menu Search

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qualitative and quantitative chemistry problems through collaborative group work. May be repeated. Must be taken Pass/D/F.

51-144 Fundamentals of Organic Chemistry

This course will assist students transitioning from general to organic chemistry, with an emphasis of making organic chemistry accessible and helping students enter the series with confidence. Students will review general chemistry concepts important to organic chemistry, including Lewis structures, acid-base chemistry, and bond polarity. Students will also learn how to name and interpret the structure of organic molecules in three-dimensional space, how structures dictate possible chemistry, and how to use spectroscopy to determine elements of chemical structure. (NS)

51-201 Chemical Methods & Techniques Lab II

Students will conduct inquiry-based experiments to enhance their understanding of kinetics, thermodynamics, equilibrium concepts and other topics. Contributes to Health Studies. Prerequisites: Chemistry 51-101 and Chemistry 51-103. To be taken concurrently with Chemistry 51-203. (NS) (Spring)

51-203 General Chemistry II

This course will introduce topics such as thermodynamics, kinetics and equilibrium. Contributes to Health Studies. Prerequisites: Chemistry 51-103 and Chemistry 51-101. To be taken concurrently with Chemistry 51-201. Students must demonstrate mastery of important concepts from the first semester of general chemistry if the course was not taken at Southwestern or during the previous semester. (NS) (Spring)

51-311 Organic Chemistry Laboratory Techniques

Students will be introduced to organic laboratory techniques with a focus on separation, purification, spectroscopy, and product analysis. Prerequisites: Chemistry 51-201 and Chemistry 51-203. Previous or concurrent registration in Chemistry 51-313 is required. (NS) (WA) (Fall)

51-313 Organic Chemistry I

 Menu Search

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connections between theory and application. Synthetic and mechanistic approaches will be introduced that lay the groundwork for Organic Chemistry II. Spectroscopic methods for structural determination of organic molecules will be discussed. Contributes to Neuroscience. Prerequisites: Chemistry 51-203 and Chemistry 51-201. (NS) (Fall)

51-321 Organic Chemistry Laboratory Application

Students will focus on the development of organic chemical reactivity, with an emphasis on synthesis and spectroscopy. Prerequisite: Chemistry 51-313 and 51-311. To be taken concurrently with Chemistry 51-323. (NS) (WA) (Spring)

51-322 Organic Chemistry Laboratory

This integrated organic chemistry laboratory course introduces students to the synthesis of molecules and the determination of reaction mechanisms. An emphasis is put on the spectroscopic characterization of reaction products and modern techniques in organic chemistry. Techniques include chromatography, distillation, extraction, melting point, and polarimetry. The curriculum is designed to make connections with culturally relevant and socially responsible topics. Prerequisite: Chemistry 51-203 and Chemistry 51-201. Previous or concurrent registration in Chemistry 51-313 is required. (NS) (WA) (Fall and Spring)

51-323 Organic Chemistry II

This course is the continuation of Chemistry 51-313. The primary focus will be on the reactions of functional groups: their mechanism and their use in complex synthesis. An introduction to biologically relevant classes of organic molecules as well as modern synthetic methods in organic chemistry will also be included. Contributes to Neuroscience. Prerequisite: Chemistry 51-313, and Chemistry 51-311 or 51-322. (NS) (Spring)

51-332 Organic Chem Lab Intermediate

The Organic Chemistry lab intermediate course focuses on advanced synthetic techniques and the characterization of organic compounds. Students engage in multistep synthetic reactions, exploring synthetically useful transformations. The course emphasizes practical

 Menu Search

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(nuclear magnetic resonance) and infrared spectroscopy. The course also emphasizes scientific communication, problem-solving in organic research, and the effective use of scientific literature. Prerequisite: Chemistry 51-322.

51-334 Synthetic & Mechanistic Organic Chemistr

This course focuses on advanced concepts in organic chemistry dealing broadly with mechanistic determination and synthetic strategies. Subtle electronic and steric factors that greatly affect the reactivity of molecules will be illustrated using a variety of case studies from current literature. These factors will then be applied to the synthesis of complex organic molecules of biological and industrial importance. Emphasis will be placed on modern methods in asymmetric synthesis and organometallics. Prerequisite: Chemistry 51-323. (NS)

51-404 Inorganic Chemistry & Biological Systems

This course is an introduction to the structure and reactivity of metal complexes with an emphasis on their interaction with biological systems. Fundamental concepts such as electronic structure, symmetry, and molecular orbital theory will be used to determine the structure of transition metal complexes. These models will then be used to describe reactivity with regard to bioinorganic chemistry. Prerequisite: Chemistry 51-313. (NS)

51-414 Organometallic Chemistry

Metal-carbon bonds lie at the interface of classical organic and inorganic chemistry. This course will explore the nature of this type of bond with an emphasis on its importance in catalysis, biological systems, and pharmaceutical drug design. Fundamental concepts such as symmetry and structure of transition metal complexes will be used to elucidate mechanistic information on the reactions of organometallic complexes. Prerequisite: Chemistry 51-323. (NS)

51-424 Fund of Materials Science & Engineering

This course focuses on the emergence of structural properties from atomic and molecular-scale interactions by conducting a survey of three broad classes of materials: metals,

 Menu Search

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polymers deployed in additive manufacturing (e.g., 3D printing) and biomaterials. Also

Physics 53-364. Prerequisites: Physics 53-154. Chemistry 51-103 is a pre- or co-requisite for this course. (Spring) (NS)

51-504 Instrumentation Envir & Biolog Analysis

This course covers the basic principles and practical applications of instrumentation used to study the environment (i.e., water, air, and soil) as well as biological phenomena.

Coursework places emphasis on fundamental techniques and the most recent advances in analytical instrumentation. Contributes to Environmental Studies. Prerequisites: Chemistry 51-313. (NS)

51-514 Quantitative Chemical Analysis

This course focuses on the basic principles of analytical chemistry and how these principles apply to chemical problems. Topics of discussion include statistical analysis in chemistry, calibration methods, chemical equilibria, and spectroscopic methods of analysis.

Prerequisites: Chemistry 51-203. (NS)

51-604 Principles of Biochemistry

This course provides an overview of the major classes of biological macromolecules (proteins, carbohydrates, lipids and nucleic acids) and their functions in cellular structure, information pathways, and enzyme catalysis, and bioenergetics. This foundation will support an introduction to metabolism and key concepts in regulation of central metabolic pathways. This course is designed for students majoring in one of the natural sciences but who do not require a two-semester course in biochemistry. This course does not fulfill the requirements for a chemistry or biochemistry major or the chemistry minor. Approved as a cellular/molecular course for use in the Biology major. Prerequisite: Chemistry 51-323. (NS) (Fall)

51-614 General Biochemistry I

This course focuses on the structure and functional interrelations of proteins, carbohydrates, lipids and nucleic acids in life processes. It is the first course of a two-

 Menu Search

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careers. Approved as a cellular/molecular course for use in the Biology major. Prerequisite.

Chemistry 51-323. (NS) (Fall)

51-624 General Biochemistry II

This course focuses on the metabolism (break-down and formation) of carbohydrates, lipids and proteins in the human body and the energy involved with such processes. It is the second course of a two-semester comprehensive sequence and is designed specifically for chemistry and biochemistry majors as well as those students interested in pursuing scientific research careers. Approved as a cellular/molecular course for use in the Biology major.

Prerequisites: Chemistry 51-604 or Chemistry 51-614. (NS) (Spring)

51-644 Special Topics in Biochemistry

This course will examine current topics in biochemistry not covered in other biochemistry courses. Prerequisites: Chemistry 51-604 or Chemistry 51-614. (NS)

51-654 Biochemistry of Nucleic Acids

A survey of nucleic acid structure and function including topics such as drug- and protein-DNA interactions, molecular recognition, DNA damage modifications and mechanisms, and DNA repair. This course also describes techniques and methods used to analyze nucleic acids. Prerequisites: Chemistry 51-604 or Chemistry 51-614. (NS)

51-694 Biochemistry of Nucleic Acids

A survey of nucleic acid structure and function including topics such as drug- and protein-DNA interactions, molecular recognition, DNA damage modifications and mechanisms, and DNA repair. This course also describes techniques and methods used to analyze nucleic acids. Prerequisites: Chemistry 51-574 (NS)

51-704 Phys Chem: Thermodynamics & Kinetics

This course focuses on the fundamental understanding and the quantitative description of chemical and biochemical processes. The course covers thermodynamics (whether processes occur) and kinetics (how fast processes occur). Chemistry majors, biochemistry majors and pre-engineering students are highly encouraged to take this course in their

 Menu Search

Apply Visit Majors & Minors

51-714 Quantum Mechanics

A detailed introduction to quantum mechanics and its applications to atoms and molecules. Also Physics 53-424. Prerequisites: Chemistry 51-203, Mathematics 52-264, and Physics 53-164. (NS)

51-822 Advanced Lab in Organic Chemistry

This course explores modern methods used in synthetic organic chemistry through participation in semester long research projects. Students will use chemical databases to develop a synthetic protocol for the formation of a molecule of medical, agricultural, or cultural significance. They will then utilize laboratory techniques such as air-free conditions, solvent purification, chromatography, and spectroscopic analysis to implement their research plans. May be repeated with changed topic with permission of instructor.

Prerequisite: Chemistry 51-332 or 51-321. (NS) (WA)

51-842 Advanced Lab in Inorganic Chemistry

This laboratory course for chemistry and biochemistry majors focuses on the reactivity, structure and synthesis of inorganic and organometallic compounds. Techniques learned cover classic inorganic synthetic methods including inert atmosphere manipulations and theoretical underpinnings of instrumental techniques. Hands-on experience on advanced laboratory instrumentation is used to characterize the compounds synthesized in this course. May be repeated with changed topic with permission of instructor. Prerequisites: Chemistry 51-332 or 51-321. Concurrent enrollment or credit in Chemistry 51-404 or 51-414 or 51-424. (NS) (WA)

51-852 Advanced Lab in Analytical Chemistry

This course introduces students to analytical chemistry through participation in a semester-long research project. Students work in groups to develop an original research question and create a proposed research plan, then carry out analytical experiments using available spectroscopic, chromatographic, and mass spectral instrumentation. The course also focuses on conducting appropriate statistical analyses of collected data and contextualizing

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Prerequisites: Chemistry 51-332 or 51-321. Concurrent enrollment or credit in Chemistry 51-504 or 51-514. (NS) (WA)

51-862 Advanced Lab in Biochemistry

A laboratory-based introduction to molecular and biochemical experimental methods and techniques that are commonly used for the analysis of biological molecules. May be repeated with changed topic with permission of instructor. When taken in combination with Chemistry 51-604 or 51-614, approved as a cellular/molecular laboratory course for use in the Biology major. Prerequisites: Chemistry 51-332 or 51-321. Concurrent enrollment or credit in Chemistry 51-604 or 51-614. (NS) (WA)

51-872 Advanced Lab in Physical Chemistry

This laboratory course will provide students an opportunity to engage in inquiry-based experiments and computation in thermodynamics, kinetics, quantum chemistry, and spectroscopy. May be repeated with changed topic with permission of instructor.

Prerequisites: Chemistry 51-332 or 51-321. Concurrent enrollment or credit in Chemistry 51-704 or 51-714. (NS) (WA)

51-911 Research in Chemistry

This is a laboratory course designed to give students hands-on experience conducting research on questions relevant to chemistry or biochemistry. Students work directly with faculty members in the department to read and analyze literature, design and perform experiments, collect and analyze data, and report the results. Students must make arrangements with individual faculty members before enrolling in this course. May be repeated.

51-912 Research in Chemistry

This is a laboratory course designed to give students hands-on experience conducting research on questions relevant to chemistry or biochemistry. Students work directly with faculty members in the department to read and analyze literature, design and perform experiments, collect and analyze data, and report the results. Students must make

51-913 Research in Chemistry may be repeated.

This is a laboratory course designed to give students hands-on experience conducting research on questions relevant to chemistry or biochemistry. Students work directly with faculty members in the department to read and analyze literature, design and perform experiments, collect and analyze data, and report the results. Students must make arrangements with individual faculty members before enrolling in this course. May be repeated.

51-922 Senior Biochemistry Capstone

Topics of this course may vary. Biochemistry students will expand their knowledge of the scientific literature, write a research paper, and present their work to peers. Biochemistry majors that have completed at least two credit hours of research (CHE51-91X) or an academic internship (CHE51-94X) will enroll in a two credit hour version of this course. Biochemistry majors that have not met the research or internship requirement will take the four credit hour version of the seminar. Prerequisite: Senior standing. (WA)

51-924 Senior Biochemistry Capstone

Topics of this course may vary. Biochemistry students will expand their knowledge of the scientific literature, write a research paper, and present their work to peers. Biochemistry majors that have completed at least two credit hours of research (CHE51-91X) or an academic internship (CHE51-94X) will enroll in a two credit hour version of this course. Biochemistry majors that have not met the research or internship requirement will take the four credit hour version of the seminar. Prerequisite: Senior standing. (WA)

51-932 Senior Chemistry Capstone

Topics of this course may vary. Chemistry students will expand their knowledge of the scientific literature, write a research paper, and present their work to peers. Chemistry majors that have completed at least two credit hours of research (CHE51-91X) or an academic internship (CHE51-94X) will enroll in a two credit hour version of this course. Chemistry majors that have not met the research or internship requirement will take the

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CONTACT